

Scaling Laws of Transient Spatiotemporal Error Bursts in Topological Quantum Error Correction

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Project report (exploratory, not peer reviewed)

Code, scripts and raw results: <https://github.com/AlanPai777/qec-burst-scaling>

Table of Contents

1. [Summary](#)
2. [Background and Question](#)
3. [Simulation Setup](#)
4. [Results](#)
 - [4.1 Baseline](#)
 - [4.2 Equal error mass, small bursts](#)
 - [4.3 Burst size relative to the code \(\$r/d\$ \)](#)
 - [4.4 Burst strength](#)
 - [4.5 Burst duration](#)
 - [4.6 A combined formula and how it compares with alternatives](#)
 - [4.7 Burst orientation](#)
 - [4.8 Does an informed decoder help?](#)
 - [4.9 Circuit-level noise](#)
 - [4.10 Long bursts with real-hardware parameters](#)
5. [What the Results Suggest](#)
6. [Limitations](#)
7. [Next Step](#)
8. [Tools and Reproducibility](#)
9. [References](#)

1. Summary

I simulated a surface-code quantum memory (code distance $d = 3$ to 9) in which a group of qubits becomes much noisier for a short time. I call this a burst. The question was whether the damage a burst does can be predicted from a few numbers that are measured relative to the code, without knowing what caused the burst.

Main results:

- **Size relative to the code matters most.** Using the burst radius divided by the code distance (r/d) predicts logical failure across $d = 5, 7, 9$ much better than the raw radius. Held-out error in $\log_{10} P_L$ is 0.29 for r/d and 0.71 for r , so a typical prediction is off by about a factor of 2 versus a factor of 5 . The result reproduces under circuit-level noise (0.285 vs 0.533).
- **Duration matters less cleanly.** Dividing duration by d helps only a little (0.78 vs 0.96). Away from saturation, failure probability grows roughly in proportion to the number of burst rounds.
- **Orientation matters a lot, and I had not planned to test it.** Two ellipses with the same qubit count and the same shape, rotated by 90° , differ in failure probability by $2.5\times$ to $21\times$, and the gap grows with code distance. A disk-based formula cannot see this. The effect also shows up for randomly grown irregular clusters (Pearson $r = 0.52$, $p \approx 0.01$, $n = 24$).
- **Real-hardware-like bursts are severe at $d = 9$.** With burst durations taken from two published events, logical failure at $d = 9$ rises from $2\text{--}4\%$ (no burst) to $30\text{--}50\%$. At $d = 5$ the background noise alone already gives $32\text{--}43\%$ over these long runs, so those rows say little about the burst (Section 4.10).

What this does not show: a single unified formula in all of size, strength and duration; anything about bursts that recur over a long computation; or any statement about the physical cause of real bursts.

2. Background and Question

Quantum error correction. A physical qubit fails often. Error correction stores one logical qubit across many physical qubits and repeatedly measures syndromes, which are parity checks that show where errors probably happened without reading the stored data. A decoder uses the syndromes to choose a correction. In the surface code with code distance d , the logical qubit uses about d^2 data qubits. If the physical error rate is below a threshold p_{th} , a larger d gives a lower logical failure probability P_L . Google's Willow processor showed this behaviour on hardware [1].

Bursts. The same paper reports rare events in which about 30 qubits become noisy together, with the extra error decaying over about $400\ \mu s$, roughly once an hour. The authors do not know the cause. In their high-distance repetition-code experiments these events set a logical error floor of about 10^{-10} [1]. An earlier experiment showed the same kind of effect: in a distance- 25 repetition code, a single high-energy event set

a floor of 1.7×10^{-6} per cycle, and excluding it gave 1.6×10^{-7} [10]. Other groups report bursts of unidentified or only partly identified origin. In one synchronized-detector experiment, cosmic rays accounted for only 17.1 ± 1.3 % of the correlated-error events [11], and a gap-engineered chip still showed frequency-shift bursts of about 1 ms with no identified cause [7].

Prior work. Tan et al. studied a burst that lasts one syndrome round and covers the entire code. They assume the decoder knows the round in which the burst happens. Under their phenomenological noise model, a burst error rate of 7 % lowers the sustainable threshold only from about 3 % to about 2.7 % [2]. They state that they did not study bursts of limited spatial extent. Sriram et al. studied whole-code bursts that recur at random over a long computation. The longest run of burst rounds grows as $O(\log L)$ with system size and dominates failure, so a 2D surface code loses its threshold if the burst error rate exceeds the bulk threshold [3]. Chadwick et al. use a finite burst radius in their noise models to detect bursts and then shut down and re-map the affected parts of a magic-state factory, which does not need to preserve logical information over time. That is an engineering strategy, not a study of how memory failure scales [4]. Kam et al. show in simulation that correlated noise can reverse the expected distance scaling even when average error rates are below threshold [12]. I did not find work that varies the spatial extent of a burst to study how memory failure scales with code size.

One reading of the hardware papers is that a whole-code burst was a fair model for small chips and is a poorer one now. In the 26-qubit experiment of McEwen et al., a cosmic-ray impact raised the number of simultaneously decaying qubits from about 4 to about 24, which saturated the chip [6]. The Willow event affects roughly 30 qubits, a neighbourhood on a larger chip [1]. This is my interpretation, not a statement from either paper.

Question. For a burst of radius r , duration τ and strength p_b in a distance- d code, does P_L depend mainly on dimensionless combinations such as r/d , τ/d and p_b/p_{th} ? I started from the guess

$$P_L \approx F(\chi), \text{ with } \chi = (r/d)^\alpha \cdot (\tau/d)^\beta \cdot g(p_b/p_{th}),$$

where α , β and the function g are to be measured. I tested this guess by trying to break it. I did not fit a flexible model to all the variables at once: such a model would fit almost any data and would not show whether a simple description exists. Fitting the exponents freely and checking whether they tie together (Section 4.6) is a light version of the dimensional-analysis approach of Bakarji et al. [14].

3. Simulation Setup

I generate rotated surface-code memory circuits with Stim [8] and decode with minimum-weight perfect matching in PyMatching [9]. The logical observable is the X-basis memory. Each configuration is sampled 8,000–100,000 times (the long runs of Section 4.10 use 8,000–10,000) and P_L is reported with a 95 % Wilson interval.

Terms used below

Term	Meaning in this report
P_L	Probability that the decoded logical observable is wrong at the end of the run
d	Code distance; $d = 3, 5, 7, 9$ (25, 49, 81 data qubits for $d = 5, 7, 9$)
p_0	Background depolarizing probability per data qubit per round (0.02 in most experiments)
Burst	In chosen rounds, the depolarizing probability of chosen data qubits is raised to p_b . No other qubits change.
r	Radius of the disk that selects burst qubits, in Stim coordinate units. Qubit spacing is 2, so r/d is a constant multiple ($2\times$) of "radius in qubit spacings divided by d ".
τ	Burst duration in syndrome rounds
p_{th}	Threshold of the background noise model. I estimate 8.5 % for the data-qubit-only model (by finding where $P_L(d)$ curves cross, the idea used by Wang et al. [15]) and 0.7 % for the circuit-level model. These are working estimates, not finite-size-scaling fits.
Ratio	p_b / p_{th}
Unaware decoder	Matching weights come from the circuit without the burst (the default in all experiments except 4.8)
Held-out RMSE	For each d in turn, hide its data, predict its $\log_{10} P_L$ from the other two distances, and take the root-mean-square error. Smaller is better.

Default parameters (each experiment overrides some of these; the table lists the values used)

Experiment	d	Rounds	Burst start / duration	p_b	Radii / other	Shots
Baseline (4.1)	3, 5, 7, 9	d	none	none	–	50,000
Equal mass (4.2)	5, 7, 9	d	mid-run / 1	0.17 ($r = 1.5$), 0.0367 ($r = 3.5$)	1 and 9 qubits	100,000
Spatial sweep (4.3)	5, 7, 9	d	round $\lfloor d/2 \rfloor / 1$	0.15	$r \in \{1.5, 3, 4.5, 6, 8, 10, 12\}$	50,000
Strength sweep (4.4)	5, 7, 9	d	mid-run / 1	ratio $\times 0.085$, ratio $\in \{0.5, 1, 2, 4, 8\}$	$r = 4.5$ (21 qubits)	50,000
Duration sweep (4.5)	5, 7, 9	24	round 4 / $\tau \in \{1, 2, 4, 8, 16\}$	0.15	$r = 4.5$ (21 qubits)	50,000
Radius $\times$ strength grid (4.6)	5, 7, 9	d	mid-run / 1	ratio $\times 0.085$	7 radii $\times$ 5 ratios = 105 cells	20,000
Ellipses (4.7)	5, 7, 9	d	mid-run / 1	0.17	equal-area ellipses, 0° and 90°	50,000
Circuit-level (4.9)	5, 7, 9	d	mid-run / 1	0.056 (ratio 8×0.007)	7 radii	50,000

Two calibration choices affect every number. The first attempt used $p_0 = 0.003$, which gave zero logical failures at $d = 5, 7, 9$ in 20,000–50,000 shots, so I raised it to 0.02. This value was chosen so that failures are countable at $d \leq 9$, not to match any specific hardware.

4. Results

4.1 Baseline (no burst)

With $p_0 = 0.02$ and no burst, P_L falls as d grows, as theory predicts: 0.93 %, 0.27 %, 0.070 % and 0.040 % for $d = 3, 5, 7, 9$. I treated this as the check needed before adding bursts.

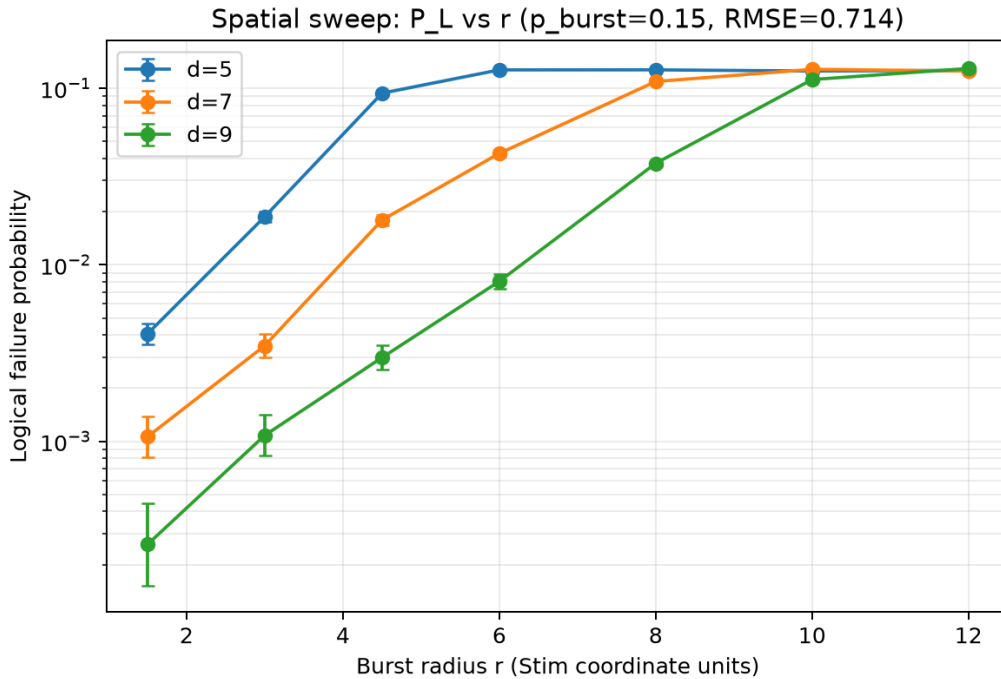
4.2 Equal error mass, small bursts

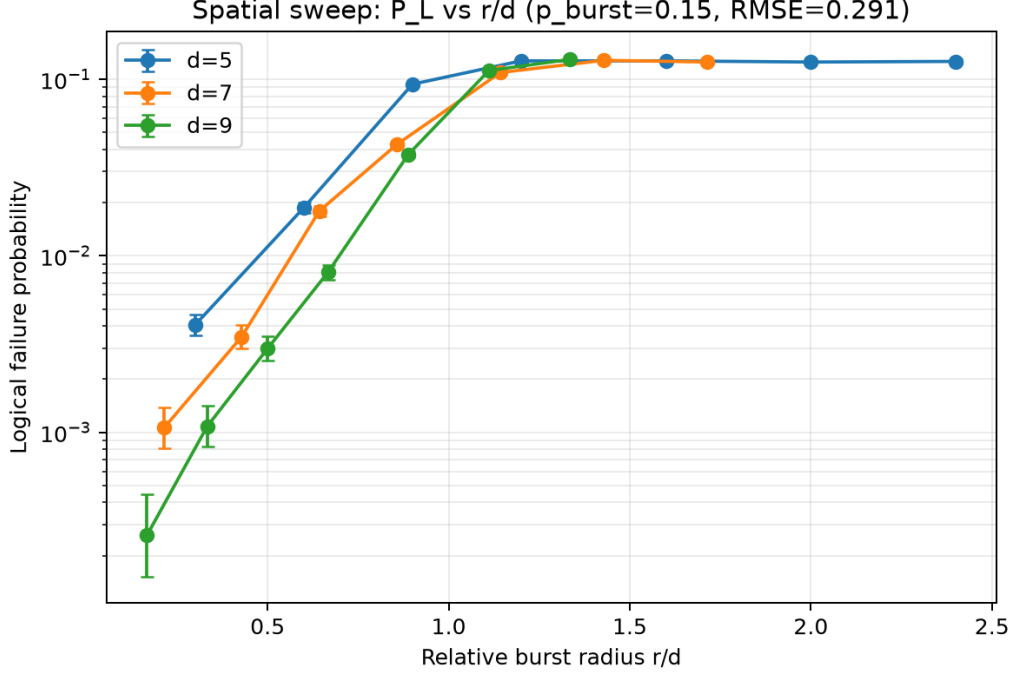
The simplest alternative to my question is that only the total extra error matters. I compared a one-qubit burst at $p_b = 0.17$ with a nine-qubit burst at $p_b = 0.0367$. Both add the same total excess, computed as (number of qubits) $\times (p_b - p_0) = 0.15$. The two give the same failure rate at every distance. At $d = 5$ the values are 0.421 % (CI 0.383-0.463) and 0.397 % (0.360-0.438).

Both bursts are small compared with the code, so this only shows that total error is enough in the small-burst regime. That result is why I next pushed the radius up to the size of the code.

4.3 Burst size relative to the code (r/d)

I used one radius list ($r = 1.5$ to 12) for all three code distances, so that r/d varies within the data instead of being fixed by construction. Strength was $p_b = 0.15$ for one round.





Figures 1 and 2. Logical failure probability against raw radius (top) and r/d (bottom) for $d = 5, 7, 9$. The curves lie much closer together in the second plot.

Held-out RMSE in $\log_{10} P_L$ is 0.71 for r and 0.29 for r/d . Once the burst covers every data qubit, P_L levels off near 0.125–0.13 at all three distances ($d = 5$, $r \geq 6$: 0.127; $d = 7$, $r \geq 10$: 0.128; $d = 9$, $r \geq 12$: 0.130). This plateau is the whole-code, one-round case that Tan et al. studied [2] and is set by the burst itself, not by d .

4.4 Burst strength

At $r = 4.5$ (21 qubits) and one round, I varied the burst strength from 0.5 to 8 times p_{th} .

Ratio (p_b / p_{th})	$d = 5$	$d = 7$	$d = 9$
no burst	0.27 %	0.070 %	0.040 %
0.5	0.65 %	0.15 %	0.018 %
1	2.8 %	0.43 %	0.076 %
2	12.1 %	2.3 %	0.40 %
4	36.2 %	10.6 %	1.9 %
8	49.9 %	21.0 %	3.7 %

Table 1. P_L against burst strength.

P_L rises smoothly and shows no jump at $p_b = p_{th}$. A burst at half the threshold raises P_L about $2.4\times$ at $d = 5$ and $2.1\times$ at $d = 7$. At $d = 9$ it is not distinguishable from the no-burst baseline, since both are at the 10^{-4} level with few failures. The same burst covers a smaller fraction of the larger code, which is consistent with 4.3.

4.5 Burst duration

For a fixed disk ($r = 4.5$, 21 qubits) at $p_b = 0.15$, I varied the burst length from 1 to 16 rounds in 24-round runs. These runs are longer than the d -round runs elsewhere, so absolute values here cannot be compared with other sections.

τ (rounds)	$d = 5$	$d = 7$	$d = 9$
1	10.4 %	1.85 %	0.35 %
2	17.7 %	3.55 %	0.56 %
4	28.7 %	6.39 %	1.18 %
8	40.3 %	11.7 %	2.27 %
16	47.7 %	20.6 %	4.43 %

Table 2. P_L against burst duration.

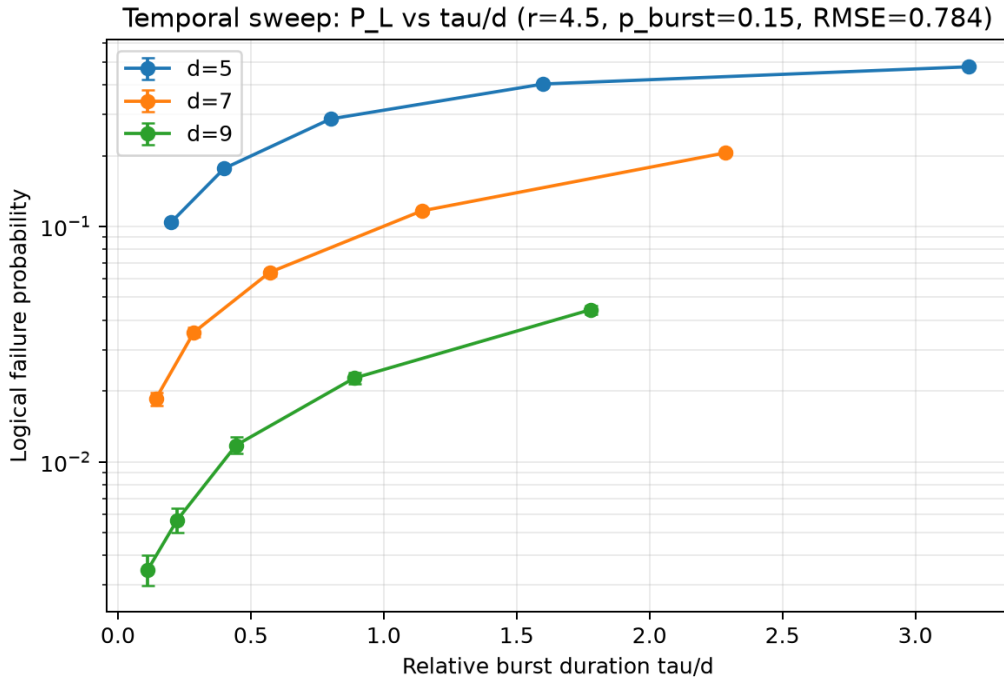


Figure 3. P_L against τ/d .

Held-out RMSE is 0.96 for τ and 0.78 for τ/d . That is an 18 % improvement, compared with 59 % for r/d in Section 4.3, so dividing by d helps much less for duration than for radius. Away from saturation, $\log P_L$ grows nearly linearly with $\log \tau$, with a slope of 0.87 at $d = 7$ and 0.94 at $d = 9$ (least-squares fit over $\tau = 1$ –16). Doubling the length of a burst roughly doubles P_L . The $d = 5$ curve saturates near 0.5, which flattens its slope to 0.56.

Radius acts far more steeply than duration (Section 4.6 gives $P_L \propto (r/d)^{2.4}$), so doubling a burst's radius does much more damage than doubling its length.

4.6 A combined formula and alternatives

I combined the grid of 7 radii $\times$ 5 strengths $\times$ 3 distances (105 cells) into one fit. I did not assume that r and d appear as r/d . I first fitted

$$\log_{10} P_L = c + a \cdot \log_{10} r + b \cdot \log_{10} d + g \cdot \log_{10}(\text{ratio})$$

with three free coefficients and checked whether $a \approx -b$. Then I compared it with the constrained two-parameter model

$$\log_{10} P_L = c + \alpha \cdot \log_{10}(r/d) + \gamma \cdot \log_{10}(\text{ratio}).$$

- **Unconstrained fit:** $a = +2.28 \pm 0.14$, $b = -3.45 \pm 0.40$, so $a + b = -1.18 \pm 0.43$. This is 2.8 standard errors away from zero, so the data reject an exact r/d tie.
- **Constrained fit:** $\alpha = 2.41 \pm 0.14$, $\gamma = 1.61 \pm 0.10$.

Model (fit on two distances, tested on the third)	Held-out RMSE ($\log_{10} P_L$)
Constrained: $(r/d)^\alpha \cdot \text{ratio}^\gamma$	0.529
Unconstrained: r , d , ratio separately	0.538
Raw radius + ratio (no d)	0.727
Total error mass only	0.803
Q3DE effective-distance formula [5]	3.107

Table 3. Held-out comparison of predictors. Cells where P_L has already saturated are included in the fit; I did not exclude them.

The r/d tie is not exact, but the constrained model predicts held-out data as well as the model with one more free parameter, so I use it as the simpler description. The radius exponent of about 2.4 is close to 2, which suggests the affected area rather than the radius may be the natural variable. I did not test that. The Q3DE formula was derived for small anomalies, and my grid goes up to whole-code coverage, so its poor score reflects a regime mismatch as much as a weakness of the formula.

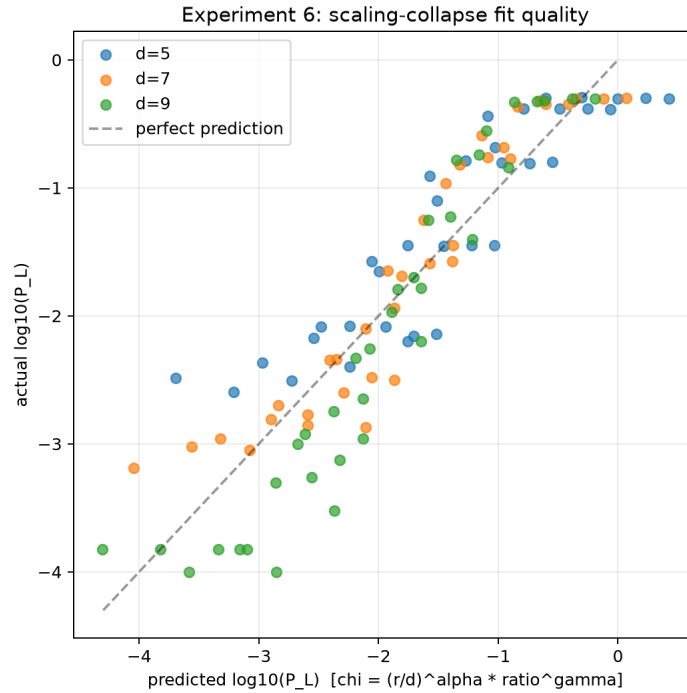


Figure 4. Constrained-formula prediction against measured P_L , coloured by d .

4.7 Burst orientation

To check whether the disk formula holds for other shapes, I built ellipses with the same area as a disk and rotated each by 0° or 90° . The two rotations contain exactly the same number of qubits, so qubit count cannot explain a difference. Strength was $p_b = 0.17$ (ratio 2), one round.

d	Disk radius target	Eccentricity	Qubits	P_L at 0°	P_L at 90°	Ratio
5	4.5	moderate	15	9.57 %	3.86 %	2.5×
5	4.5	high	5	2.06 %	0.57 %	3.6×
5	6.0	moderate	15	9.55 %	3.76 %	2.5×
5	6.0	high	5	2.07 %	0.58 %	3.5×
7	4.5	moderate	17	3.42 %	0.68 %	5.0×
7	4.5	high	7	0.76 %	0.14 %	5.5×
7	6.0	moderate	21	5.83 %	0.80 %	7.3×
7	6.0	high	7	0.89 %	0.16 %	5.5×
9	4.5	moderate	19	1.43 %	0.17 %	8.6×
9	4.5	high	9	0.32 %	0.03 %	10.5×
9	6.0	moderate	27	3.86 %	0.18 %	21.4×
9	6.0	high	9	0.37 %	0.03 %	10.8×

Table 4. Same-shape, same-qubit-count ellipses differing only by a 90° rotation. The 90° cells for the high-eccentricity ellipses at $d = 9$ rest on 15–17 failures in 50,000 shots, so those ratios are uncertain by roughly a factor of two.

The ratio grows with d in every group. The disk-only fit of Section 4.6 predicts the same value for both rotations, so it cannot see this effect.

Why the direction matters. I inspected the compiled circuit and found that the tracked logical X operator runs along a single line of qubits. The dangerous orientation (0°) is perpendicular to that line and the safe one (90°) is parallel to it. A burst stretched across the operator's line gives errors a shorter route between the two boundaries that the logical operator joins. This is a percolation-style picture, and I have not tested it beyond the orientation data itself. Hastings et al. show, for higher-dimensional codes at finite temperature, that a memory can survive beyond the percolation threshold of its defects [13], so a connected region of noisy qubits would not by itself prove failure. I treat the picture as a hypothesis.

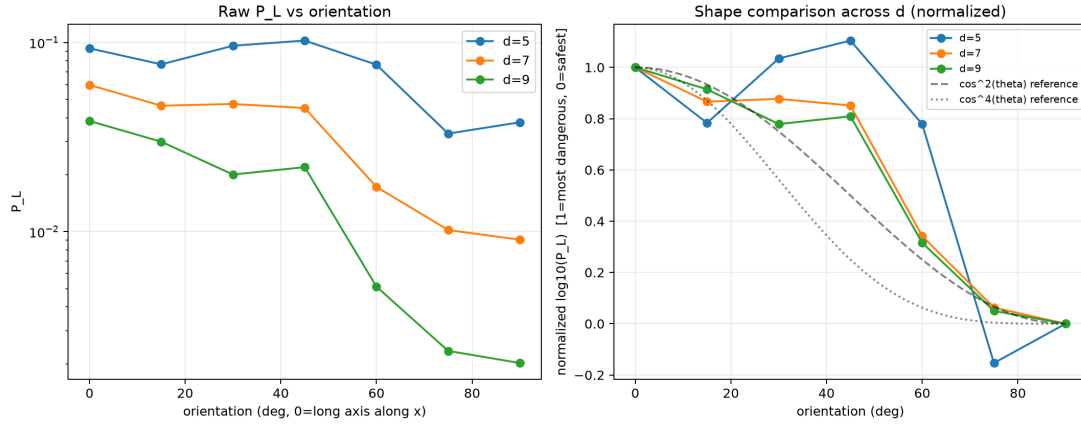


Figure 5. P_L against burst orientation, normalised within each d (1 = most dangerous angle, 0 = safest).

Angle sweep. I swept the angle from 0° to 90° in 15° steps for the moderate-eccentricity ellipse (disk radius target 6). At $d = 9$, P_L is 3.8 %, 3.0 %, 2.0 % and 2.2 % from 0° to 45° , then falls to 0.51 % at 60° , 0.23 % at 75° and 0.20 % at 90° . This is a step-like drop between 45° and 60° – 75° , not a smooth $\cos^2$ curve. The qubit count changes with angle on the discrete lattice (25–31 at $d = 9$), so only the 0° and 90° endpoints are cleanly controlled.

Random shapes. To check that the effect is not specific to ellipses, I grew 24 irregular 21-qubit clusters (8 random seeds $\times$ 3 distances) and compared the disk formula's error with a "directional signal" for each cluster: (eccentricity) $\times$ $\cos(2 \times \text{angle to the dangerous axis})$, from a principal-component fit. The correlation is $r = 0.52$ ($p \approx 0.01$). This is moderate, and the 24 points are not fully independent because the same seeds are reused at each d .

4.8 Does an informed decoder help?

Tan et al. report that a decoder told about the burst does about as well as one that is not [2], but they only tested whole-code bursts. I reran the 105-cell grid with matching weights built from the circuit that includes the burst. The average difference in $\log_{10} P_L$ is near zero. It grows to +0.18 (about $1.5\times$) at the strongest bursts (ratio 8) and is +0.13 for bursts with r/d below 0.73, and it is about zero when the burst nearly covers the whole code. For the small and strong bursts that resemble real hardware events, telling the decoder helps.

4.9 Circuit-level noise

The experiments above use noise on data qubits only. I added depolarizing noise after gates, measurement flips and reset flips (all at 0.003, with $p_{th} \approx 0.7$ % found by curve-crossing) and repeated the radius sweep at ratio 8 ($p_b = 0.056$). To do this I had to fix how the injection code found round boundaries: it originally counted `DEPOLARIZE1` instructions, which breaks once gate noise adds more of them, so it now counts `TICK` instructions (7 per round). I checked that the fix reproduces the earlier results exactly.

Held-out RMSE is 0.285 for r/d and 0.533 for r . The values for data-qubit-only noise were 0.291 and 0.714, at a different strength. So the advantage of r/d holds under circuit-level noise.

4.10 Long bursts with real-hardware parameters

I ran two published burst scenarios, using a conversion of 1 μ s per syndrome round. Willow's cycle time is 1.1 μ s [1], so this is reasonable for that chip. I have no source for the other chip, so treat its duration as approximate:

- **Willow-like** [1]: ~ 30 qubits (an ellipse, 15–27 qubits at $d = 5$ –9), exponential decay with a 400-round time constant, 800 rounds.
- **Kurilovich-like** [7]: a disk of radius 6 (25–29 qubits, size chosen by me because the paper gives none), 1000-round time constant, 1500 rounds.

Neither source gives the burst's error probability, so I set the peak strength to 1, 2 and 4 times p_{th} and the strength decays exponentially. I also ran the same circuits without any burst, because over hundreds of rounds the background noise alone produces logical failures.

Scenario	d	No-burst control	Peak ratio 1	Peak ratio 2	Peak ratio 4
Willow-like	5	31.6 %	49.5 %	49.7 %	50.1 %
Willow-like	9	1.9 %	30.5 %	49.5 %	50.4 %
Kurilovich-like	5	43.1 %	49.9 %	49.9 %	50.1 %
Kurilovich-like	9	4.1 %	34.4 %	50.1 %	50.2 %

Table 5. P_L for long, decaying bursts. 50 % means the decoder has lost all information.

At $d = 5$ the control is already 32–43 %, so the near-50 % values there are mostly background noise and say little about the burst. The $d = 9$ rows are the informative ones. A burst at the threshold raises P_L from 2–4 % to 30–34 %, and any stronger burst takes it to about 50 %. I did not extend the fit from Section 4.5 to these durations. In units of τ/d they are 14–35 times larger than the largest value I tested (3.2).

Given this, a single such burst in the middle of a memory experiment at $d = 9$ is very likely to cause a logical failure. What is not answered is how often bursts happen compared with the length of a computation, which is the question in Section 7.

5. What the Results Suggest

These are suggestions that follow from the simulations. I have not tested any of them on hardware.

- **Where to spend mitigation effort.** In this model, a burst's footprint relative to the code (r/d , exponent about 2.4) and its orientation change P_L far more than its length does (P_L roughly proportional to τ). Reducing the affected area, or moving it away from the direction that crosses the logical operator, would help more than shortening the burst by the same factor.

- **Telling the decoder.** For small, strong bursts an informed decoder reduced P_L by up to about $1.5\times$ (Section 4.8). Q3DE [5] and ReloQate [16] already react to detector statistics, but with hand-chosen thresholds. A rule based on the estimated footprint relative to d would be a possible alternative.
- **What hardware papers could report.** Papers on correlated errors usually report a duration and a rate. My results suggest that the number of affected qubits, their layout relative to the logical operator, and the code distance are also needed to judge how serious an event is.

6. Limitations

- **Formula is incomplete.** The fit covers radius and strength. It has no orientation term and no duration term. Duration was measured separately with a different round count, so I could not include it in the same fit.
- **Saturation.** The fit in Section 4.6 includes cells where P_L is near its ceiling. A power law is not expected to hold there, and I did not check how much the exponents change if those cells are dropped.
- **Idealised noise.** Most results use depolarizing noise on data qubits only, at $p_0 = 0.02$, chosen for statistics. Circuit-level noise (Section 4.9) checks the r/d result but not the others.
- **Small codes.** I simulated $d \leq 9$, so extrapolating to larger codes is untested.
- **One burst, one place, one decoder.** Each run has one burst, in the middle of the run, decoded by matching. The decoder-aware comparison (4.8) is the only test of a different decoder.
- **Unit conversion.** The $1\ \mu\text{s}$ per round conversion in Section 4.10 is only sourced for Willow. Neither published burst gives a strength directly, so strength was swept.
- **Orientation explanation.** The account of why direction matters (Section 4.7) is consistent with the data but is not tested independently.
- **Statistics.** The strongest orientation ratios come from cells with only 15–17 failures.

7. Next Step

The main open question is what happens when bursts recur. The results here suggest one burst is severe at $d = 9$ (Section 4.10), so what matters is how often bursts occur and how large they are. Sriram et al. show that for whole-code bursts a 2D surface code has no decodable phase once the burst error rate exceeds the bulk threshold [3]. It is not known whether limiting the burst to a finite radius restores a threshold.

Testing this needs a different simulation from the one used here: bursts triggered at random each round over runs much longer than d rounds, several code sizes to look for a crossing point, and orientation included, since Section 4.7 shows it changes the outcome. I have not started it. A first step would be to reproduce the whole-code result of Sriram et al. in my pipeline before adding a radius.

8. Tools and Reproducibility

Every number above comes from a script in `experiments/` and a timestamped CSV in `results/`. Runs use fixed random seeds, Stim 1.16.0 and PyMatching 2.4.0. The repository is at <https://github.com/AlanPai777/qec-burst-scaling>.

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